Predicting Daily Calories from Google Fit Data

SE ZG568 Applied Machine Learning | Assignment EC-1 | Personal Data Story

Real Google Fit export (Motorola Edge 40 Neo) | Patna, Bihar | 638 days | Aug 2024 – May 2026

PART 1 — Data & Problem Framing

Data Provenance

Data was automatically recorded by Google Fit on my **Motorola Edge 40 Neo** smartphone using built-in accelerometer and activity recognition. Downloaded via **Google Takeout** (takeout.google.com) on **4 May 2026** — selecting only the Fit category, which exported JSON files for 5 data streams: step count delta, calories expended, active minutes, distance, and heart minutes.

Transformations:

JSON records aggregated to daily totals. Timestamps converted from nanoseconds UTC to IST (UTC+5:30). Days with <200 steps (phone not carried) excluded. Days >80,000 steps (sensor glitch) removed. **First 62 rows (Jun–Jul 2024) excluded** — systematic calorie underestimation due to missing biometric profile on newly set-up phone (see Quirk 3).

Three Verbatim Raw Records:

```
2024-08-01 | steps=25983 | cal=1854.8 | active=201min | dist=10.89km | hm=10.0 | wkday
2024-12-31 | steps=27088 | cal=1894.8 | active=206min | dist=12.32km | hm=85.0 | wkday
2025-09-07 | steps=12082 | cal=2165.2 | active= 68min | dist= 3.31km | hm= 4.0 | wkend
```

Problem Statement

Target: calories (kcal/day, continuous)
Type: Supervised Regression
Modules: Regression (Module 4) + Ensembles (Module 6)

Relevance: Predicting daily calorie expenditure from easily measurable phone signals helps me estimate energy burn on days when sensors partially fail, enabling proactive meal planning for personal weight management.

Data Dictionary

Feature	Type	Unit
steps	int	count/day
active_min	int	minutes
distance_km	float	km
heart_min	float	points
is_weekend	int	0/1
month	int	1–12
log_steps *	float	—
intensity_ratio *	float	pts/min
calories †	**float**	**kcal**

** Engineered feature   † Target variable*

Three Domain-Specific Quirks

Quirk 1 — Motorola Batch-Sync Artifact:

The Edge 40 Neo uses aggressive battery-saving batch sync. Steps near midnight are sometimes attributed to the next IST calendar day, creating ~5–10% step misattribution between adjacent days — a timing noise pattern absent from continuous-sync devices like Fitbit/Garmin.

Quirk 2 — Bimodal Step Distribution (College Days vs. High-Activity Days):

Most weekdays cluster at 22,000–35,000 steps (campus commuting), while weekends/holidays push above 50,000 steps (treks, travel, sports). ~8% of days exceed 50k steps. This bimodality differs from typical office-worker data and means a single linear model cannot fit both regimes.

Quirk 3 — Calorie Underestimation in First Two Months (Jun–Jul 2024):

First 62 days show 700–1,400 kcal despite high step counts, because Google Fit had not yet received height/weight profile on the newly set-up phone. Mean early calories ~1,200 kcal vs. post-calibration mean ~2,075 kcal. These 62 rows were excluded from modeling.

Three Pre-Modeling Challenges

• Challenge 1: steps, active_min, and distance_km are near-perfectly correlated (r>0.90). I expect multicollinearity to inflate OLS coefficients and motivate Ridge regularization.

• Challenge 2: The bimodal step distribution means two distinct behavioral regimes. I expect linear models to underfit both and tree-based models to handle splits between regimes better.

• Challenge 3: High-activity days (>50k steps, ~8% of data) are sparse in training. I expect the model to underestimate calories on these extreme days due to too few examples.

PART 3 — Modeling: Iteration Log & Final Model

#	Model	Key Reasoning	Val MAE	Val R^2
1 *Baseline*	OLS Linear Regression	Simplest baseline. OLS showed multicollinearity problem: steps/active_min/distance_km produced unstable competing coefficients. R^2 ~0.40 — surprisingly low given strong EDA correlations.	195.7	0.398
2	Ridge Regression (α=0.001)	L2 regularization to address multicollinearity. Best α=0.001 (near-zero) shows regularization magnitude is not the core issue — linearity assumption is. Only marginal improvement.	194.7	0.396
3 FINAL	**Random Forest (n=500, depth=None, leaf=4, feat=log2)**	**Non-linearity fix: recursive splits handle bimodal step distribution. Random feature subsampling resolves multicollinearity structurally. 50-iter RandomizedSearchCV, 5-fold CV. Large R^2 jump confirms non-linearity was root cause.**	**140.9**	**0.669**

Feature Engineering (3 features): (1) **log_steps** — log-transforms right-skewed step count; diminishing calorie returns at high steps. (2) **intensity_ratio** = heart_min/(active_min+1) — distinguishes a brisk run from a slow walk of equal steps. (3) **active_hour_frac** = active_min/1440 — normalized fraction of day active, scale-independent.

Final Model Justification: Random Forest (n=500, max_features='log2', min_samples_leaf=4) was selected because: (a) R^2 improved from 0.40 (OLS) to 0.67 (RF) on validation — a 68% relative gain — proving non-linearity was the bottleneck, not multicollinearity; (b) feature importances align with domain expectations (active_min and steps dominate); (c) 5-fold CV used for hyperparameter selection rules out validation-set overfitting.

PART 4 — Evaluation & Error Analysis

Primary Metric: MAE (kcal)

From a diet-planning perspective, being off by 200 kcal is twice as costly as being off by 100 kcal — I under/over-eat proportionally to the error. This is a linear, symmetric cost function, making MAE the correct primary metric. RMSE is also reported to penalize the large extreme-day errors that would disrupt a multi-day plan. R^2 gives a scale-free fit summary. Accuracy is inapplicable.

Test Set Performance (96 held-out days: 28 Jan – 4 May 2026)

Model	Test MAE (kcal)	Test RMSE (kcal)	Test R^2	vs Baseline
Baseline: OLS Linear Regression	246.4	359.3	0.333	—
Attempt 2: Ridge (α=0.001)	245.0	357.3	0.340	MAE –1.4
FINAL: Random Forest	**229.2**	**332.0**	**0.430**	**MAE –17.2 R^2+0.098**

Baseline vs Final: The Random Forest improves MAE from 246.4 to 229.2 kcal and R^2 from 0.333 to 0.430 on the held-out test set. Both models perform worse on the test period (Jan–May 2026) than on validation — this test window covers an exam period with fewer high-activity outdoor days and a denser cluster of anomalous low-calorie days (sensor glitches, partial days near export date). The RF still consistently outperforms OLS across all three metrics; the added complexity of 500 trees is justified by the directional consistency of improvement.

Error Analysis — 5 Worst Prediction Cases

#	Date	Key Inputs	True	Pred	Err	Hypothesised Cause

1	2026-03-11 Wed	steps=57,624 active=388min hm=98	1,433	2,510	–1,076	Motorola batch-sync artifact (Quirk 1): steps from previous night misattributed to this date, inflating step count. True calorie label is correct but input is noisy. Model prediction is reasonable for the (inflated) step count.
2	2026-03-22 Sat	steps=34,200 active=309min hm=51	3,278	2,231	+1,047	Missing feature: likely cycling or uphill trekking session — activities with high calorie burn but moderate steps and heart_min. Model cannot distinguish activity type from steps alone. Structurally missing feature.
3	2026-03-01 Sat	steps=15,909 active=127min hm=46	788	1,803	–1,015	Noisy label: calorie value of 788 kcal is below realistic BMR — Google Fit calorie stream had a data gap. Likely ill or phone not carried for most of day. Model prediction is plausible; true label is suspect.
4	2026-05-04 Mon	steps=9,185 active=68min hm=29	804	1,712	–909	Truncated-day record: May 4 is the Google Takeout export date (05:26 IST). Only a few hours of data recorded before export. Both steps and calories are partial-day values — out-of-distribution input the model cannot detect.
5	2026-05-03 Sun	steps=33,707 active=338min hm=119	3,049	2,261	+788	High-intensity terrain activity: high heart_min (119) relative to steps suggests hilly trek or sports with significant elevation. Training data has very few examples of this combination. Model has correct direction (high hm → high cal) but underestimates magnitude.

Challenges Revisited

Challenge 1 (Multicollinearity): CONFIRMED — OLS showed competing coefficients. Ridge's best α=0.001 (near-zero) proved regularization magnitude wasn't the issue. RF's subsampling resolved it structurally. **Challenge 2 (Non-linearity/bimodal):** CONFIRMED — R^2 jump from 0.40 to 0.67 on validation directly quantifies the non-linearity gap. **Challenge 3 (Sparse high-step regime):** CONFIRMED — Error Cases 2 and 5 show underprediction on high-calorie days; Cases 1, 3, 4 show sensor glitches/partial days producing anomalously low true labels.